Half-coalescence and reset renewal on a record-atom tail

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Let

$$x_{t+1} = U_+(x_t) := \frac{3x_t + 1}{2^{a_t}}, \qquad a_t = v_2(3x_t + 1), \qquad A_t = \sum_{h<t} a_h, \tag{1}$$

be an injective infinite accelerated orbit of positive odd integers. Put

$$Q_t := \frac{3^t}{2^{A_t}}, \qquad 0 < \mathcal{K} := \lim_{t\to\infty} \frac{x_t}{Q_t} < \infty, \tag{2}$$

and assume that $Q_t \to \infty$. Let $j_0 < j_1 < \cdots$ be consecutive coefficient tail records. On an interior deficit-four tail, every record gap is one of the atoms

$$A := 1, \qquad B := 12, \qquad C := 1113, \qquad D := 1122. \tag{3}$$

Write

$$b_i := v_3(x_{j_i} + 1). \tag{4}$$

The decoder transport at record endpoints is

$$A : b \mapsto b + 1, \qquad B : b \mapsto 0, \qquad C : b \mapsto 1, \qquad D : b \mapsto 0. \tag{5}$$

The synchronized companion is the sequence determined by

$$2^{a_t} e_{t+1} = 3e_t - 1, \qquad e_t > 1, \qquad \frac{e_t}{Q_t} \longrightarrow 0.$$

Adding its recurrence to (1) shows that

$$\frac{x_t + e_t}{Q_t}$$

is constant; the boundary condition and (2) identify this constant as $\mathcal{K}$. Thus $x_t + e_t = \mathcal{K} Q_t$ at every state.

Theorem 1 (exact half-coalescence at the atom C)**.** *Suppose that a record transition starts at x and has exponent word $C = 1113$. Put*

$$y := U_+^4(x). \tag{6}$$

Then

$$x \equiv 79 \pmod{128}, \qquad y = \frac{81x + 65}{64}. \tag{7}$$

The smaller positive odd integer

$$\boxed{z := \frac{x-1}{2}} \tag{8}$$

has exact accelerated exponent word 1121 *and coalesces with x at the same endpoint:*

$$\boxed{U_+^4(z) = y.} \tag{9}$$

Let $e_x, e_y > 1$ be the synchronized companion values at the two record endpoints. Then

$$64e_y = 81e_x - 65. \tag{10}$$

The alternative companion start

$$e_z := \frac{e_x + 1}{2} > 1 \tag{11}$$

is transported to e_y by the same alternative word 1121*, and*

$$\boxed{z + e_z = \frac{x + e_x}{2}.} \tag{12}$$

If x belongs to the unbounded orbit of a globally least positive odd unbounded seed n, then

$$\boxed{x \geq 2n + 1.} \tag{13}$$

If Q is the coefficient at x, so that $x + e_x = \mathcal{K}Q$, then

$$\boxed{Q > \frac{2(n+1)}{\mathcal{K}}.} \tag{14}$$

Proof. The affine carry of the word 1113 is 65 and its total exponent is 6, which gives the second identity in (7). Since y is odd,

$$81x + 65 \equiv 64 \pmod{128}.$$

Since $81^{-1} \equiv 49 \pmod{128}$, it follows that $x \equiv -49 \equiv 79 \pmod{128}$, proving the first identity in (7).

Starting from $z = (x-1)/2$, the four states generated by the proposed word 1121 are

$$\frac{x-1}{2}, \quad \frac{3x-1}{4}, \quad \frac{9x+1}{8}, \quad \frac{27x+11}{32}, \quad \frac{81x+65}{64}. \tag{15}$$

The congruence $x \equiv 79 \pmod{128}$ shows directly that the extracted powers of two are respectively $1, 1, 2, 1$. The last state in (15) is y, proving (8)–(9).

Iteration of the negative companion recurrence through 1113 gives (10). The word 1121 has total exponent 5 and carry 73, so its companion equation is

$$32e_y = 81e_z - 73. \tag{16}$$

Substitution of (11) into the right side of (16), followed by (10), proves (16). Equation (12) is immediate from (8) and (11).

The orbit of z reaches y and then follows the unbounded tail, so it is unbounded. Global leastness of n gives $z \geq n$, and (8) proves (13). Finally, $e_x > 1$ and $x \geq 2n+1$ imply

$$\mathcal{K}Q = x + e_x > 2(n+1),$$

which is (14). □

Theorem 2 (exact reset renewal on a C-free tail)**.** *Assume that, from some record onward, no atom C occurs. A* reset *is a record immediately following an atom B or D. There are infinitely many resets. Between two consecutive resets the atom word is uniquely of one of the forms*

$$A^k B, \qquad A^k D, \qquad k \geq 0. \tag{17}$$

Let x and x' be the states at two consecutive resets, and put

$$Y := x + 1, \qquad Y' := x' + 1, \qquad \nu := v_2(Y), \qquad u := \frac{Y}{2^\nu}. \tag{18}$$

Then u is odd, $3 \nmid Y$, $3 \nmid Y'$, and the following two exact alternatives hold:

reset block	ν	Y'	Q'/Q
$A^k B$	$k+2$	$\dfrac{3^\nu u + 1}{2}$	$\dfrac{3^\nu}{2^{\nu+1}}$
$A^k D$	$k+3$	$\dfrac{3^{\nu+1} u + 7}{8}$	$\dfrac{3^{\nu+1}}{2^{\nu+3}}$.

(19)

If i is the record index of the first reset in (19), then, with $\alpha = \log_2 3$,

$$\boxed{\nu \le (\alpha - 1)i + \log_2(\mathcal{K}Q_{j_0}).} \tag{20}$$

Proof. By (5), the atoms B and D end at decoder depth zero, so every reset satisfies $3 \nmid Y$. An eventual tail of atoms A alone would give an infinite exact exponent-one run. This is impossible because the exact run from a positive odd state has length $v_2(x+1) - 1$. Hence B or D occurs infinitely often, proving that there are infinitely many resets and that consecutive resets have one of the forms in (17).

The raw exponent words in (17) are

$$1^{k+1}2, \qquad 1^{k+2}22, \tag{21}$$

respectively. The exact initial exponent-one runs therefore have lengths $k+1$ and $k+2$. The exponent-one tower formula gives $\nu = k+2$ in the first case and $\nu = k+3$ in the second.

Across $A^k B$, the equations for $Y = x + 1$ give

$$2^{k+3}Y' = 3^{k+2}Y + 2^{k+2}. \tag{22}$$

Substitute $Y = 2^{k+2}u$ and divide by 2^{k+2} to obtain the first state formula in (19). The coefficient multiplier is

$$\left(\frac{3}{2}\right)^k \frac{9}{8} = \frac{3^{k+2}}{2^{k+3}},$$

which is the first multiplier in (19).

Across $A^k D$, the corresponding equation is

$$2^{k+6}Y' = 3^{k+4}Y + 7\,2^{k+3}. \tag{23}$$

Substitution of $Y = 2^{k+3}u$ proves the second state formula. Its coefficient multiplier is

$$\left(\frac{3}{2}\right)^k \frac{81}{64} = \frac{3^{k+4}}{2^{k+6}},$$

which proves the remaining entry in (19). The endpoint is again a reset, so $3 \nmid Y'$.

Finally, $2^\nu \mid Y$ and $Y < \mathcal{K}Q_{j_i}$. Consecutive record multipliers are at most $3/2$, hence

$$Q_{j_i} \le Q_{j_0} \left(\frac{3}{2}\right)^i.$$

Taking binary logarithms proves (20). □

Corollary 3 (coalescing versus two-map renewal)**.** *Every eventual four-atom interior tail has at least one of the following two properties.*

1. *The atom $C = 1113$ occurs infinitely often. Every such occurrence produces the exact half-size coalescing seed in (8), and on a globally least unbounded ray it can occur only after the coefficient threshold (14).*

2. *The atom C occurs only finitely often. After its last occurrence, the orbit has infinitely many decoder-zero resets governed exactly by the two renewal laws (19), subject to the valuation budget (20). Its atom word still satisfies the fixed-seed return-lattice and factor-complexity bounds for an interior record-atom tail.*

Proof. The two alternatives distinguish infinite from finite occurrence of C. Theorem 1 proves the first case. After the last occurrence, Theorem 2 supplies the reset renewal.

The fixed-seed return-lattice theorem applies to every interior four-atom tail and gives the final assertion. □